

CSCI3170

Introduction to Database Systems

1-entity_relationship_model
Slides: Michael Yu

Motivation/ The Business Scenario

The start of every database design project:

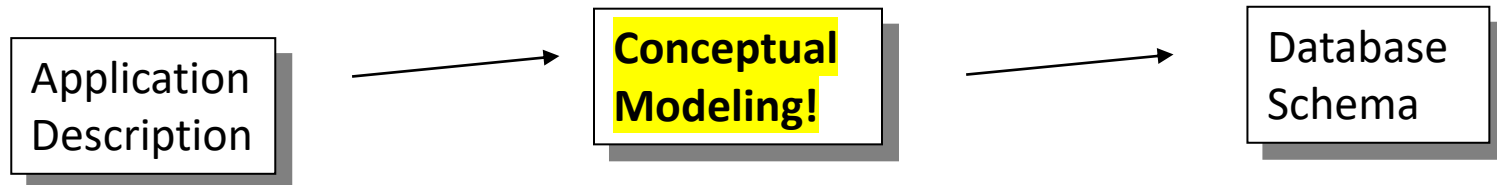
- A: “Welcome, build us a database to organize all the information on a publisher and present it?”
- **B: “okay, what’ll be in the database?”**
- A: “Well... authors, books, editors, printers etc....
- A: “An author can write zero or more books and a book is written by one or more authors.”
- A: “A book is uniquely identified by its book-id. For each book, we also record its title, price, and availability.”
- A: “An editor is uniquely identified by his/her reader-id and we also record his/her name, phone number, address. The address is composed of street and suburb...”

THE Business Application

- Basically... from a Business POV, there is an application we need to design for, and this application has a set of requirements
 - A **text-based description** of the data related needs
- Example:
 - “An author can write zero or more books and a book is written by one or more authors. Where a book is uniquely identified by its book-id. For each book, we also record its title, price, and availability. An editor is uniquely identified by his/her reader-id and we also record his/her name, phone number, address. The address is composed of street and suburb...”
- Person may or may not be an familiar with databases at all.

The Entity-Relationship Model

- Chen's ER model has two major components:
 - **Entity**: collection of attributes describing object of interest
 - **Relationship**: association between entities (objects)
- There's also the third components unofficially
 - **Attribute**: data item describing a property of interest
 - **So far, attribute we cover stores one value (atomic)**
 - sure it can change... but one value at a time

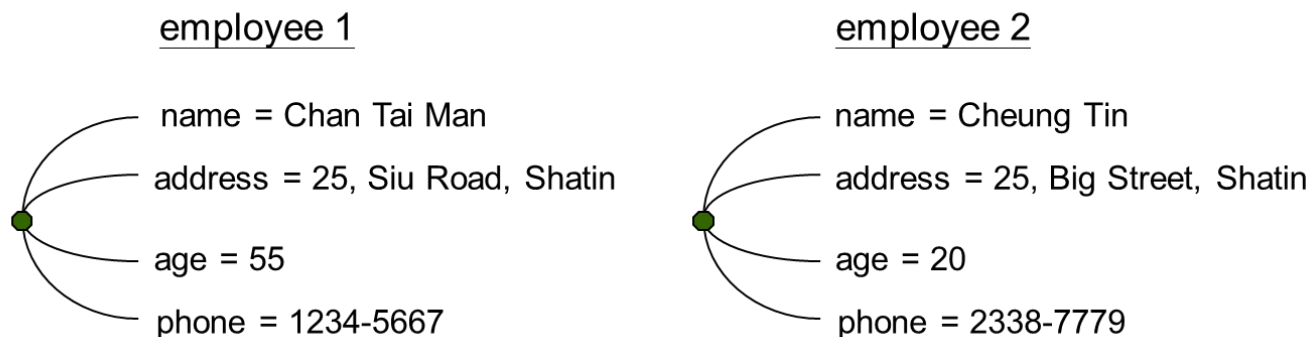


(Chen. 1976)

- Entities
- Relationships
- Attributes

Constraints on Data (First of Many)

- We have seen so far how to model a slice of the real world using entities and some (atomic) attributes. However, there are some other important aspects to the real world that we cannot model with the tools seen so far.
 - In an entity set, we need to be able to distinguish each entity
- In databases, there are constraints on the data that go beyond structural constraints.
 - To support operations, every entity must have a final **key**
 - **Keys are attributes or sets of attributes that uniquely identify an entity within its entity set.**



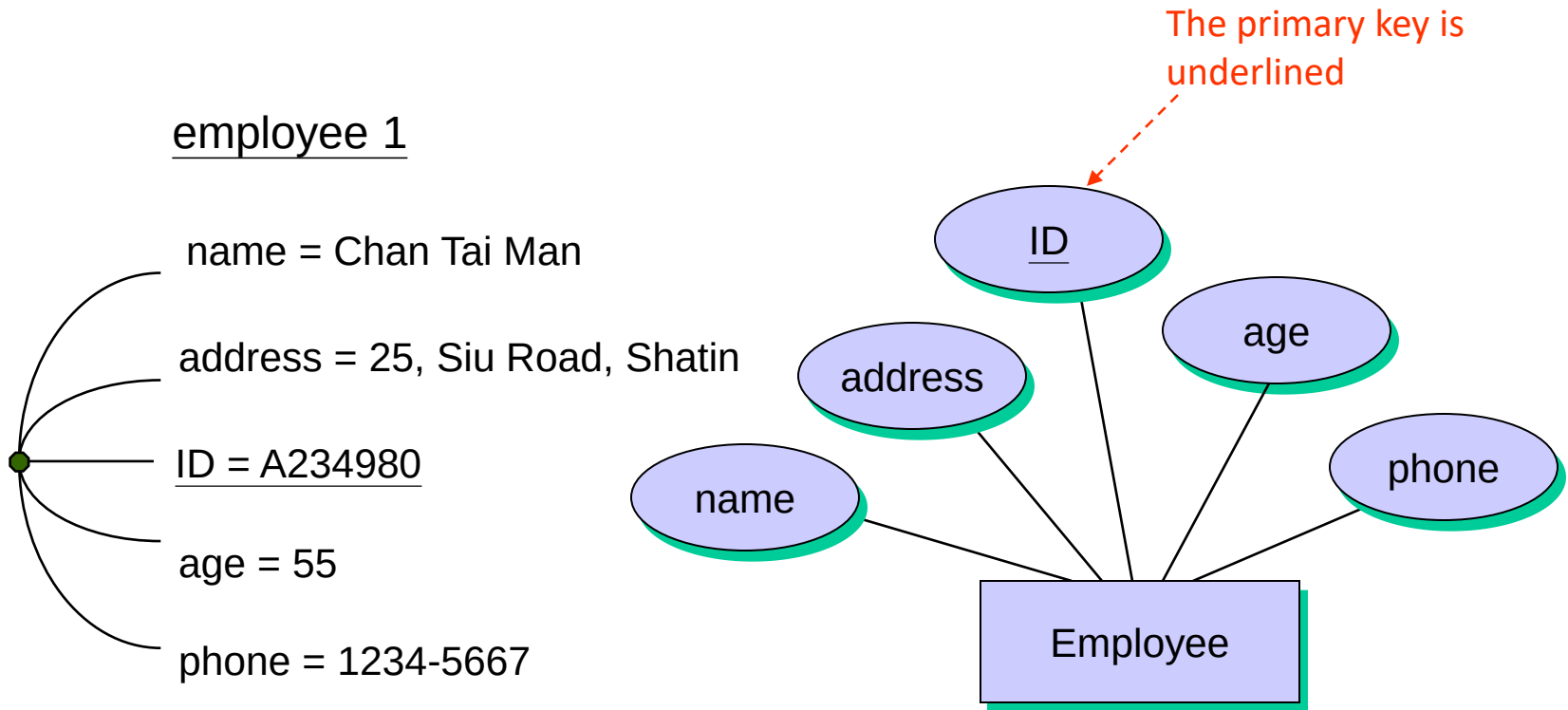
The Spirit of E/R Models

- Attributes should be properties of entities that do exist (if a value exists), and kept to those **inherent** to the entity.
 - Intrinsic attributes describe *what the entity is*, not how it relates to other things.
- **Natural Key**
 - a column or set of columns that already exist in the model/ table (e.g. they are attributes of the entity within the data model) and uniquely identify a record in the table.
 - these columns are attributes of the entity they obviously have “business meaning”.

FirstName	LastName	BirthDate	SSN	AddressID	HireDate
Jim	Smith	1978-03-21	234-24-2343	1	2010-02-17
Tammy	Jones	1994-09-21	346-73-1944	2	2011-04-27
Sam	Donald	2001-08-17	487-14-3329	5	2014-12-07
Mike	MacDonald	1999-06-28	654-81-8273	3	2009-02-11
Stephanie	Shepherd	1998-06-02	834-94-9941	6	2013-01-01
Jennifer	Bowles	1994-12-01	943-91-4733	4	2015-12-22

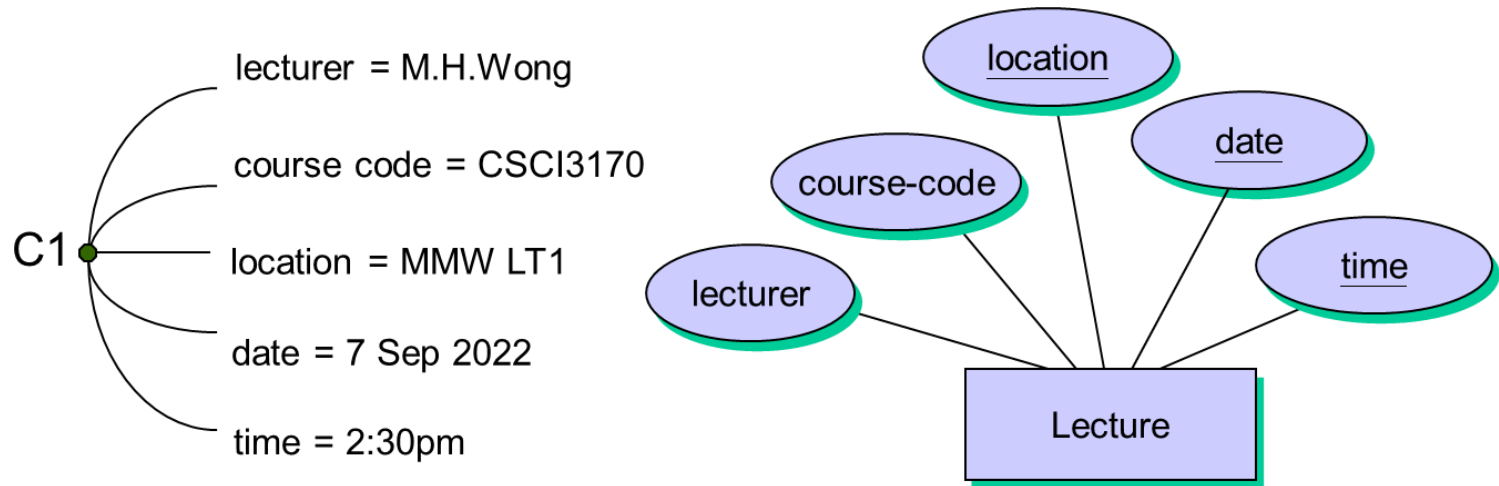
Resolution of Employee Entity (type)

- Usually, we need to add some attribute(s) as a key.



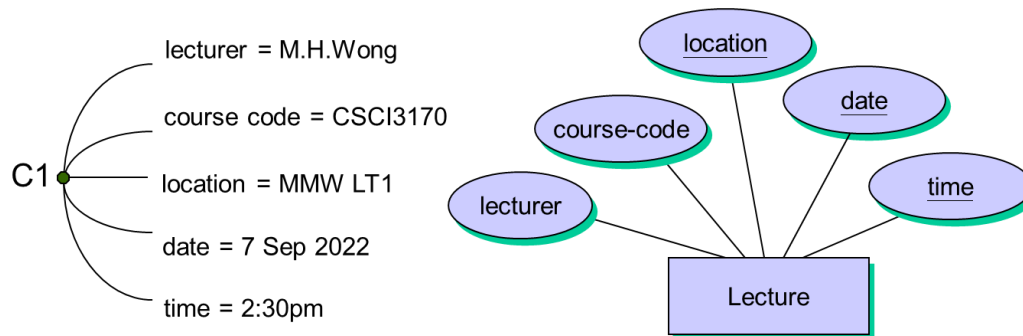
Key

- If a key has more than one attribute, then it's also known as a **composite key** or a compound key.
- A **candidate key** is a **minimal** set of attributes whose values uniquely identify an entity in the set.
 - For example, {location, date, time} is a candidate key.



More than One Key

- Q: In real-life, there is likely more one valid possible key?
 - There could be more than one candidate key.
 - A **primary key** is a candidate key chosen to serve as the key for the entity set.
- A **super-key** is any set of attributes which can uniquely identify an entity (always)
- For example,
 - {location, date, time} is a key.
 - {lecturer, location, date, time} not candidate key, is a super key.



Summary: Key Terminology

- A **super key** is any set of attributes which can uniquely identify an entity
- A **candidate key** is a minimal set of attributes whose values uniquely identify an entity in the set
 - No proper subset of the attributes can be a super key
- There could be more than one candidate key
- A **primary key** is a candidate key chosen to serve as the key for the entity set

Sometimes Practiced: Surrogate Key

- Q: Suppose no one single, or set of attributes, can uniquely identify this entity?
 - Perspectives on adding a surrogate key field?
- **Surrogate Key (not in spirit of ER)**
 - system generated (could be GUID, sequence, unique identifier, etc.) value with no “business meaning” that is used to uniquely identify a record in a table.
 - E.g., *AddressID* is a surrogate key from below

AddressID	StreetNumber	StreetName	City	State	ZipCode	ModifiedDate
1	123	6th St.	Melbourne	FL	32904	2018-04-11 20:02:41.637
2	71	Pilgrim Avenue	Chevy Chase	MD	20815	2018-04-11 20:02:41.647
3	70	Bowman St.	South Windsor	CT	06074	2018-04-11 20:02:41.647
4	4	Goldfield Rd.	Honolulu	HI	96815	2018-04-11 20:02:41.647
5	44	Shirley Ave.	West Chicago	IL	60185	2018-04-11 20:02:41.650
6	514	S. Magnolia St.	Orlando	FL	32806	2018-04-11 20:02:41.650

Look Around You

	Class Title	Enrolled	Days & Times	Room	Class Dates
CSCI 3170--T01 (7026)	Intro to Database Systems (Interactive Tutorial)	84	Mo 11:30AM - 12:15PM	Lady Shaw Bldg LT1	1/9, 8/9, 15/9, 22/9, 29/9, 6/10, 13/10, 20/10, 27/10, 3/11, 10/11, 17/11, 24/11
CSCI 3170--T02 (9167)	Intro to Database Systems (Interactive Tutorial)	13	Tu 2:30PM - 3:15PM	Y.C. Liang Hall G04	2/9, 9/9, 16/9, 23/9, 30/9
			Tu 2:30PM - 3:15PM	Y.C. Liang Hall G04	14/10, 21/10, 28/10, 4/11, 11/11, 18/11, 25/11
CSCI 3170--T03 (7423)	Intro to Database Systems (Interactive Tutorial)	28	We 4:30PM - 5:15PM	Li Koon Chun Hall LT1	3/9, 10/9, 17/9, 24/9
			We 4:30PM - 5:15PM	Li Koon Chun Hall LT1	8/10, 15/10, 22/10
			We 4:30PM - 5:15PM	Li Koon Chun Hall LT1	5/11, 12/11, 19/11, 26/11

Simple Attributes Revisited

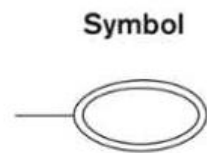
- Attributes are what you want to store about your data
 - Most common type is a simple attributes
- Each simple attribute of an entity type
 - stores one value **at a time (atomic)**
 - sure it can change... but one value at a time.
 - For example:
 - Entity = Student
 - Attributes = Student number, name...

Multivalued Attributes

- Context (Clothing Manufacture)
 - What if... I'm asking you to model shirts with two colours?
 - What properties/attributes might require multiple values?



- In E/R, we may model such attributes as **multivalued attributes**
 - Graphically denotated as double lined ovals



Meaning

Multivalued Attribute

Critical Thinking

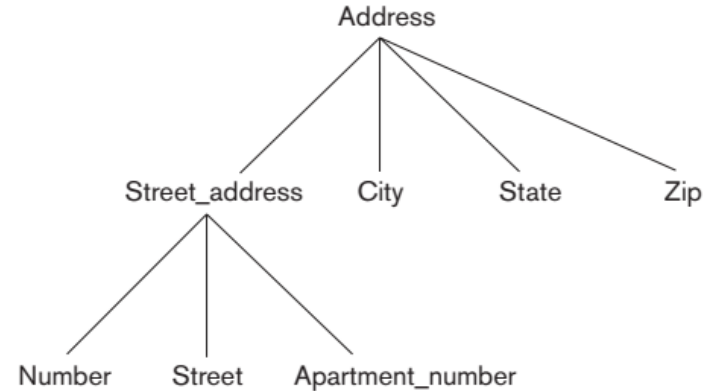
- Can we do without multi-value attributes?
 - Sometimes, **yes**.
- The limitation of E/R and only atomic attributes, is that some **semantics** cannot be captured
 - (meaning it'll be highly inefficient without contradicting the existing constraints)
 - Modelling Constraint: Simple attributes hold one value at a time
- Recall Design Principle: “Attributes you design should be able to describe, and the combinations of all its instances should be able to express the entity **faithfully**” faithfully = good.
 - Example: enables expressive modelling the attribute colour for any future shirt entity

Composite Attributes

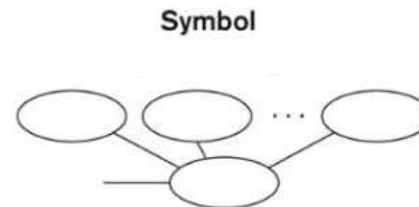
- There are more semantics cannot be captured using simple/atomic attributes (without causing some headaches later)
 - As singular/ atomic values, they are assumed to be non-divisible
 - Intuitively, there is no reason to divide a value into smaller values
 - Where possible, basic functionalities will not cater to this
- Some properties/ values do have smaller “parts”, each with it’s own independent meanings (capable as attributes)
- Consider Name as a simple attribute
 - E.g., Name value = “Michael”

Composite Attributes

- Address = 'Computer Science Building (K17), Engineering Rd, UNSW Sydney, Kensington NSW 2052'

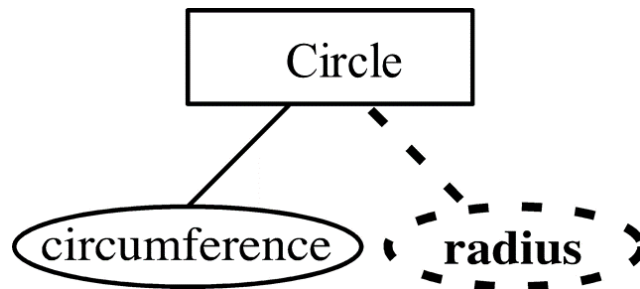


- Composite attributes are useful for situations when this property is one where an end-user, is going to ,at times, refers to the whole “property” as a complete unit, and at other times, just by its specifics components.
- Denoted graphically as a tree-like structure with child attributes



Derived Attributes

- Sample Scenario: modelling a person's age:
 - If an entity stores attributes *Date-of-birth* and *Age*
 - Age attribute should be **derived attribute**
 - Date-of-birth attribute as non-derived (stored) attribute
- Denoted graphically as a dotted-line oval.

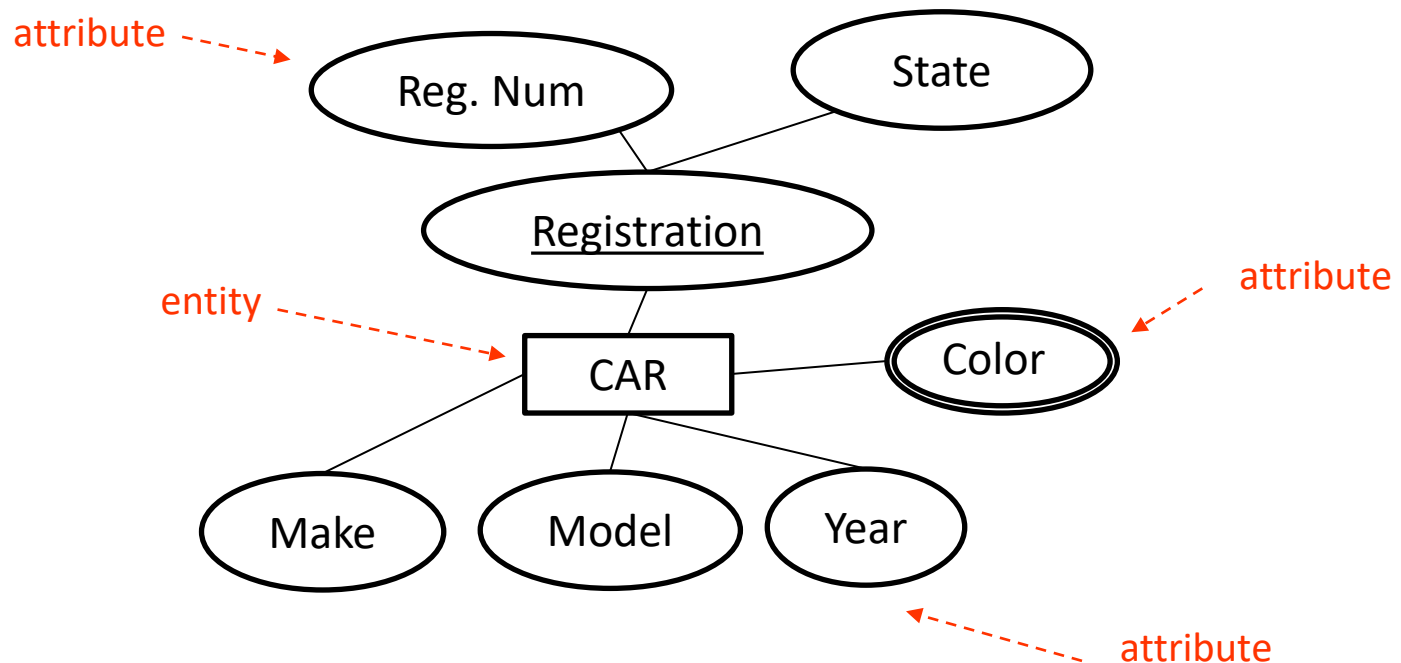


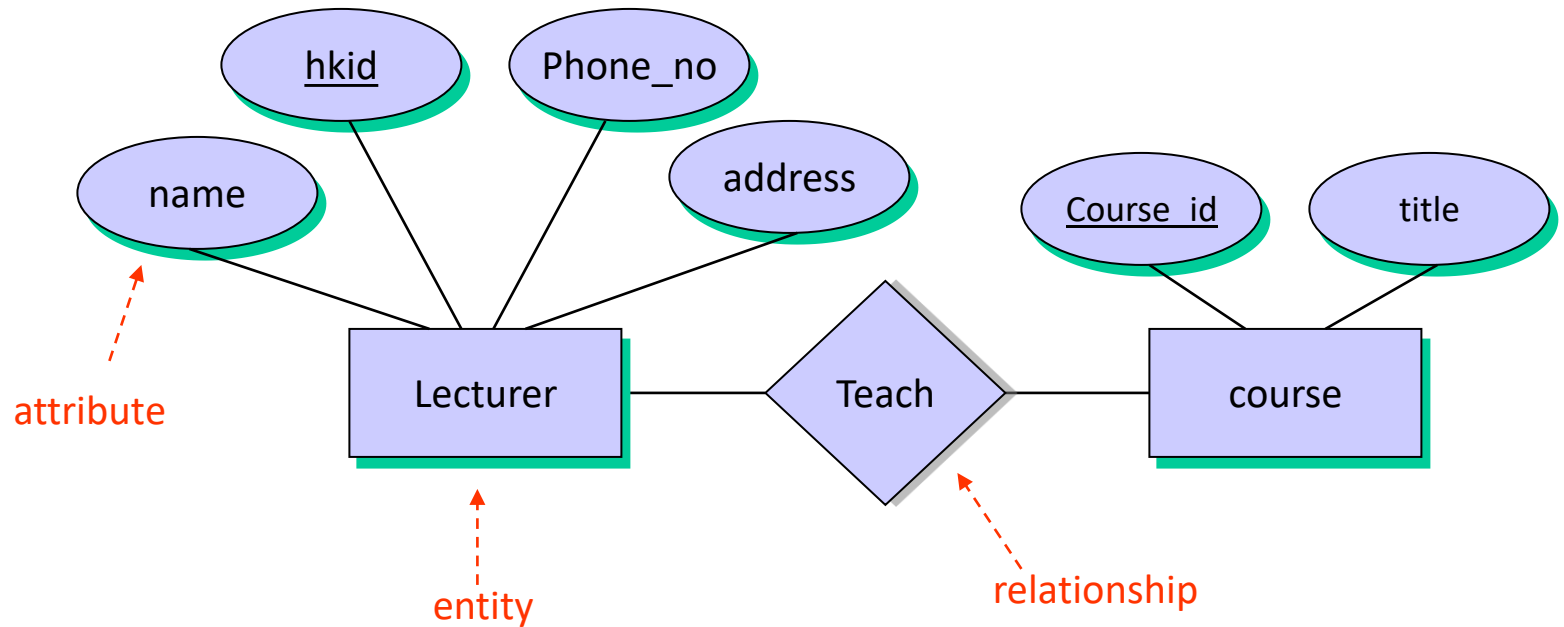
Scenario: It's nicer with E-R Diagrams

- A: “so far we completed that high level data model that you wanted sir...” ***hands you piece of paper***
- You look at paper:
 - “
 - One (strong) entity type that is a Car.
 - One multi-labelled attribute to describe its colour.
 - One composite attribute *Registration* with is made of *Registration Number* and *State*.
 - Other attributes *Make*, *Model* and *Year*
 - ...
 - ”

Scenario: It's nicer with E-R Diagrams

- “There is one (strong) entity type that is a Car. It has a multi-labelled attribute to describe its colour. It has Registration with is made of Registration Number and State. It has a few more attributes Make, Model, Year ...”





Next:

ON RELATIONSHIPS IN E-R DIAGRAMS

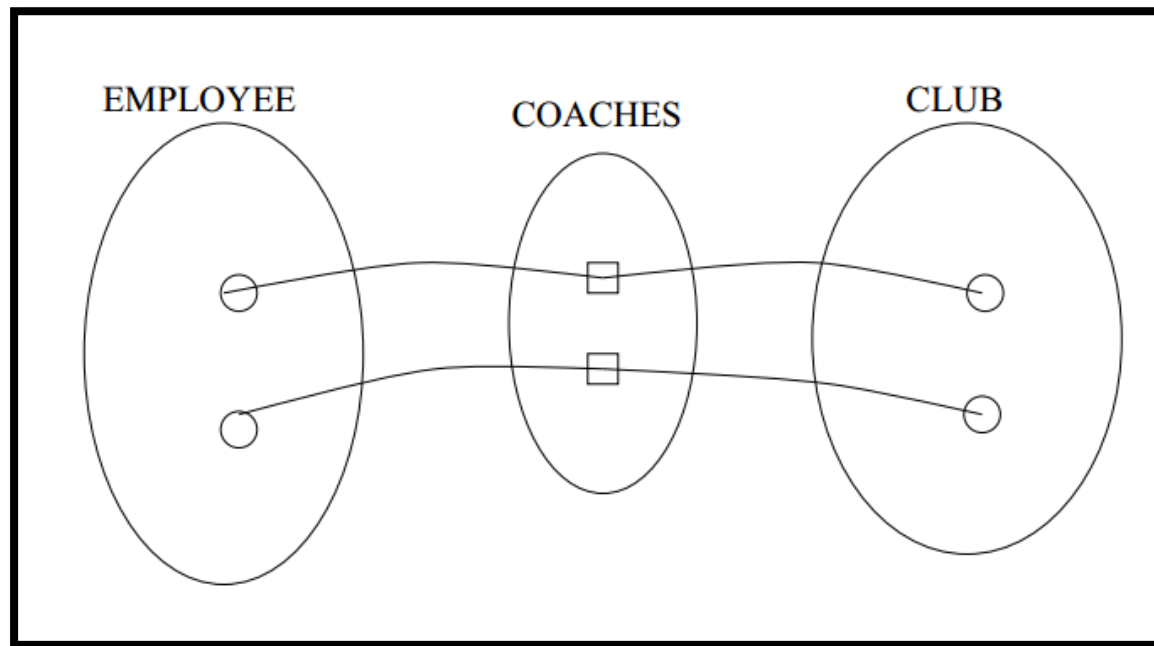
Relationship

- Entities are independent, can be stand alone. In real-world most of **what we want to store** are related.
- “relationships” are the component that allow us to model logical links between two or more entities.
- Entity A can have several different relationships with another Entity B.
 - More generally, any set of entities can are not limited to participating in at most one relationship with each other.



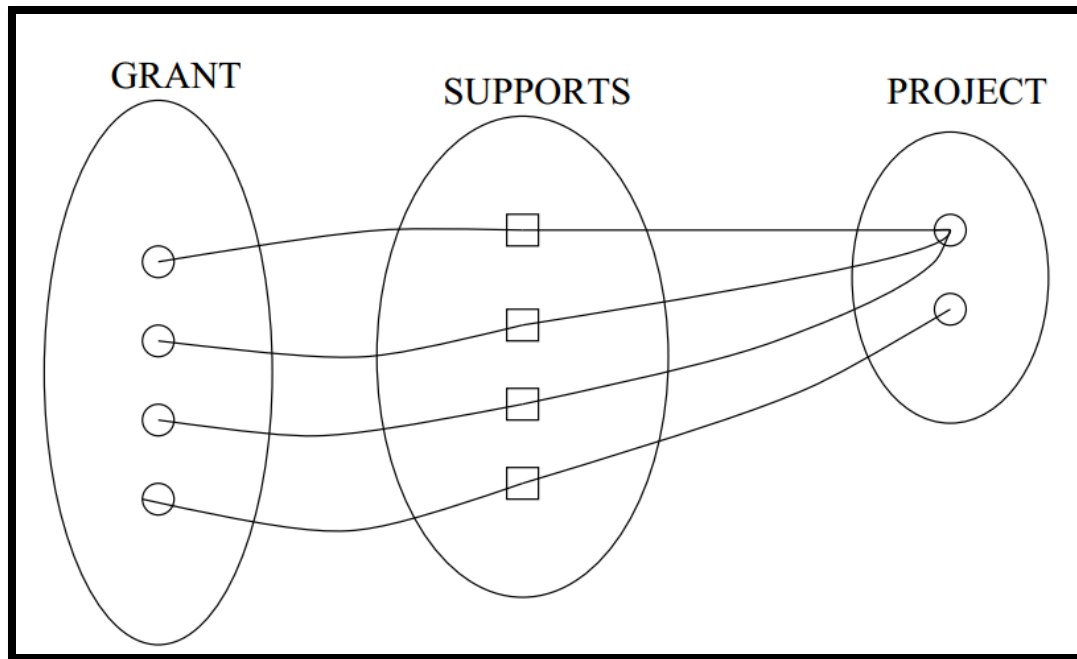
Intro to Relationships (1)

- Using a **entity-occurrence diagram**, we explore possible relationships patterns between occurrences of a particular entity.
 - **Coaches** is a relationship. It's instances describes the relation between a coach and a club.



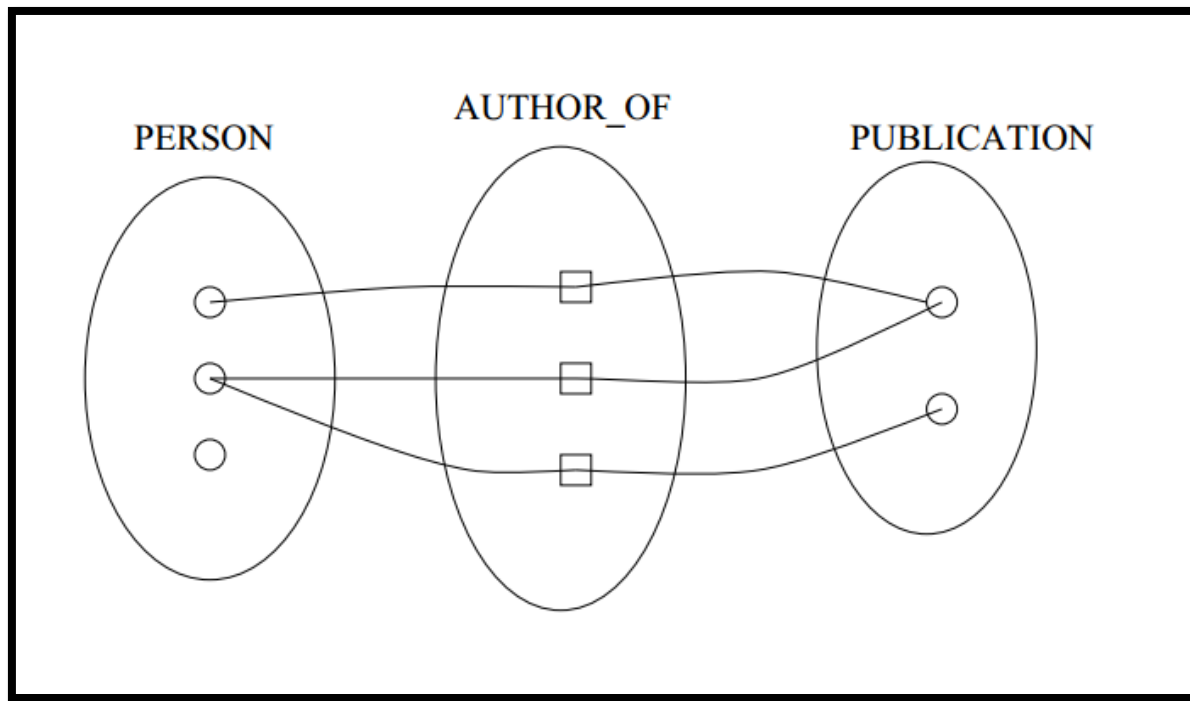
Example of Relationships (2)

- Using a **entity-occurrence diagram**, we explore possible relationships patterns between occurrences of a particular entity.
 - **Supports** is a relationship. It's instances describe the how grants are given to projects.



Example of Relationships (3)

- Using a **entity-occurrence diagram**, we explore possible relationships patterns between occurrences of a particular entity.
 - **Author-of** is a relationship. It's instances describe who wrote what publication.

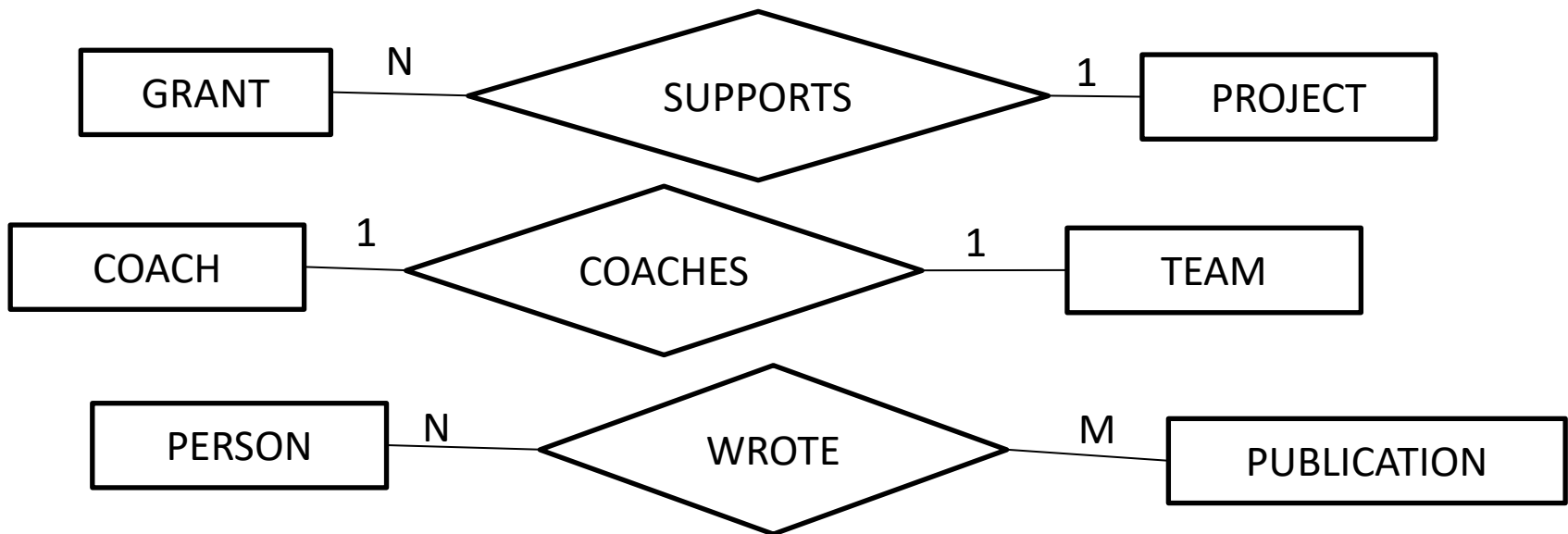


Modelling Relationships

- Not all relationships are the same, there is a need to express them faithfully. There are two constraints we apply for relationships.
 - *Cardinality*
 - *Participation*
- **Cardinality** (ratio) :
 - A constraint on the number of relationship instances an entity can participate in.
 - A relationship should be modelled as one of the following ratios:
 - M:N (many to many) least strict
 - 1:N (one to many)
 - 1:1 (one to one) most strict

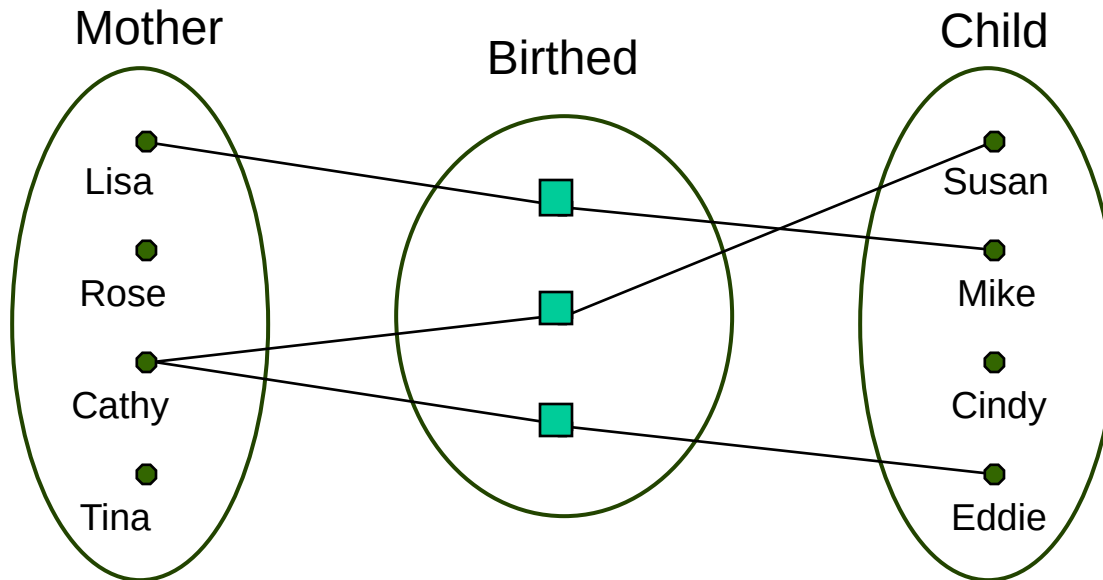
Relationship Constraints

- There are generally these kinds of real relationships, categorized by their different “cardinalities”:
- We can also show this in an E/R diagram
 - (notation below a helpful **starting practice** guideline, not to be confused our mandatory course notation)



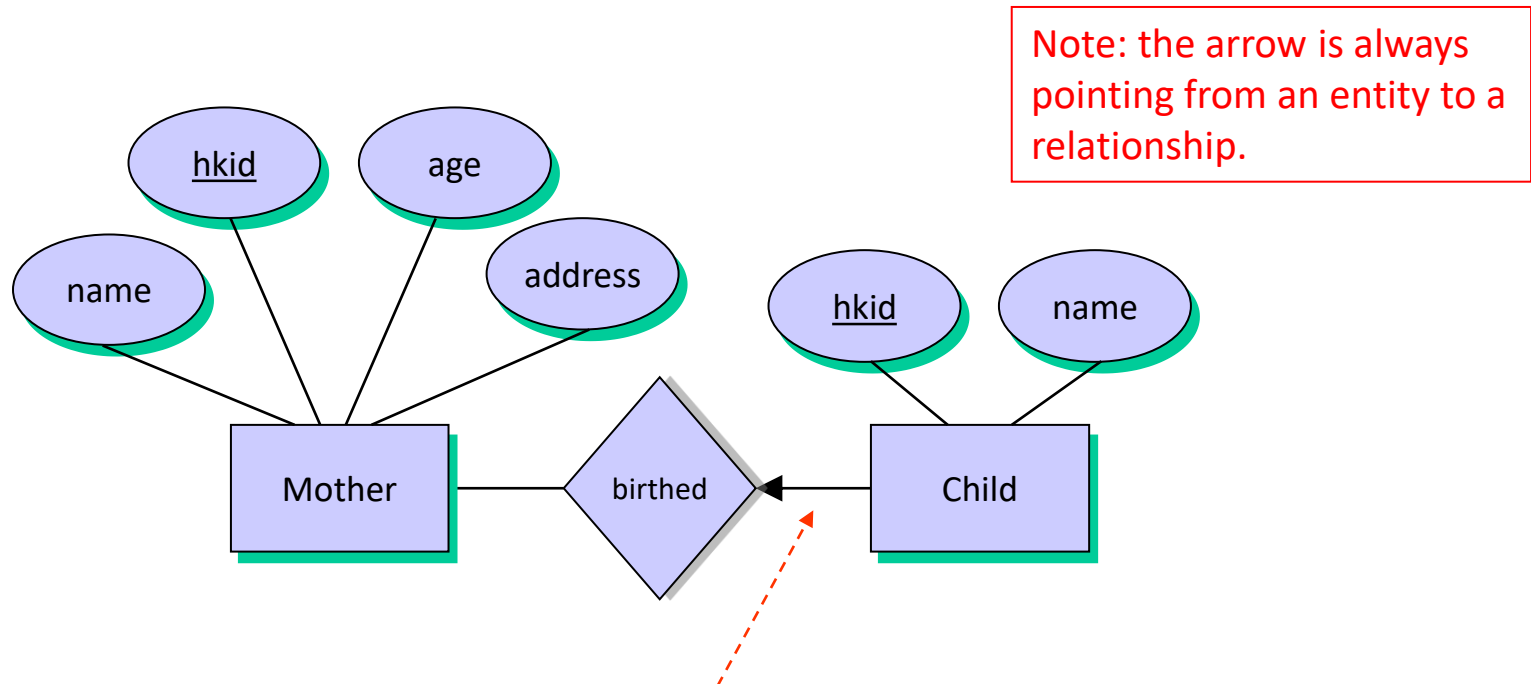
One-to-Many (Card. Constraint)

- One-to-many constraint from A to B:
 - An entity in B can be associated with **at most** one entity in A
 - *E.g., One child appear in **at most one** mother-child relationship instance.*



CSCI3170 Notation: One-to-Many

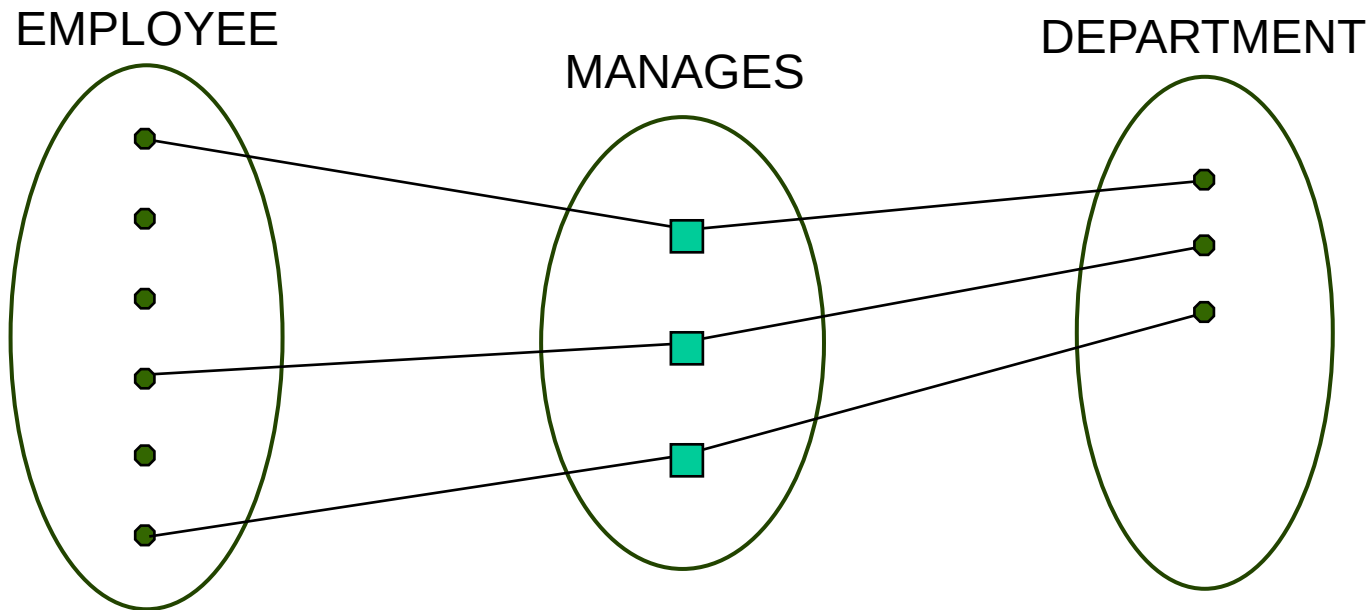
- *(pardon any political correctness sensitivities)*
 - Child has a constraint in the mother-of relationship set.
 - Restriction can be indicated by an arrow in the E-R diagram.



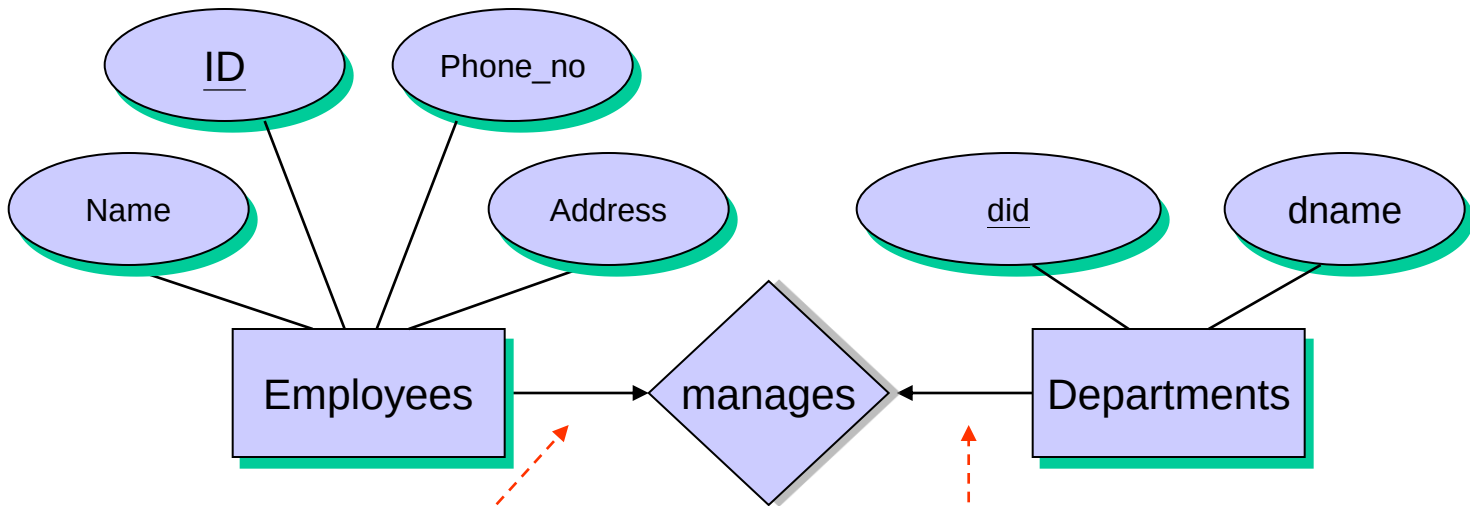
Intuitively, the arrow states that given a child entity, (if it participates in the birthed relationship), we can uniquely determine the mother-of relationship.

One-to-One (Card. Constraint)

- If relationship between A and B satisfies the one-to-one mapping constraint from A to B
 - an entity in A is related to at most one entity in B,
 - an entity in B is related to at most one entity in A.



Notation: One-to-One/1:1

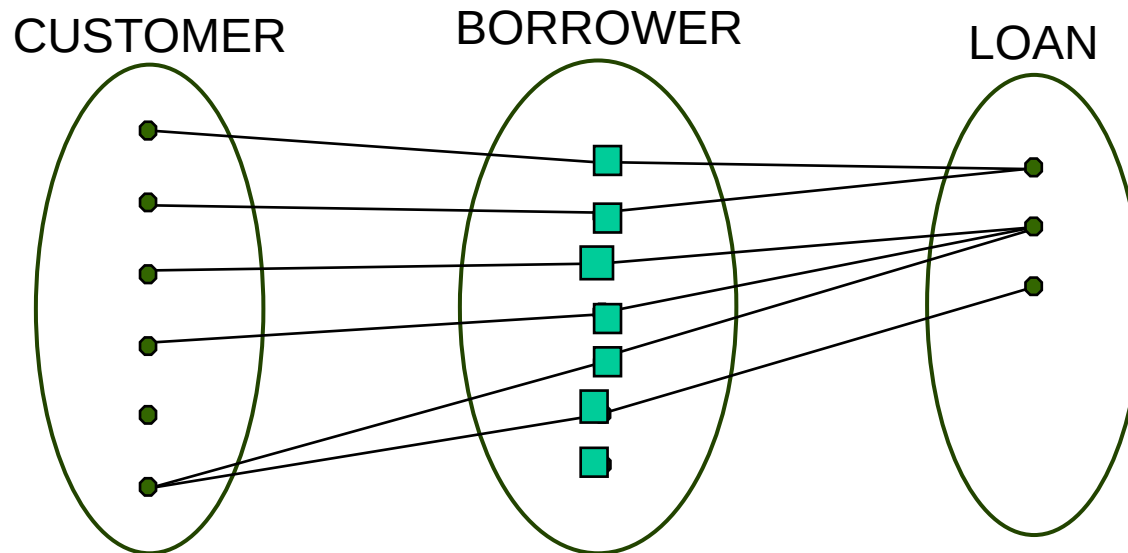


An employee can associate with **at most** one department via the relationship “manages”.

A department can associate with at most one employee via the relationship “manages”

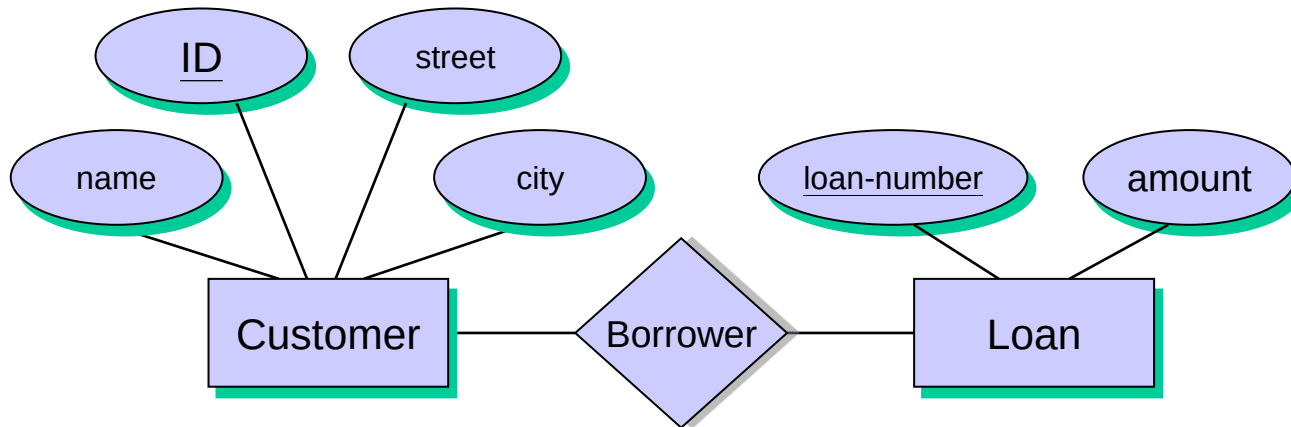
Many-to-Many (Card. Constraint)

- Many-to-many
 - An entity in A is associated with **any number** of entities in B,
 - An entity in B is associated with **any number** of entities in A.



- i.e., **no** restriction in relationship set membership.

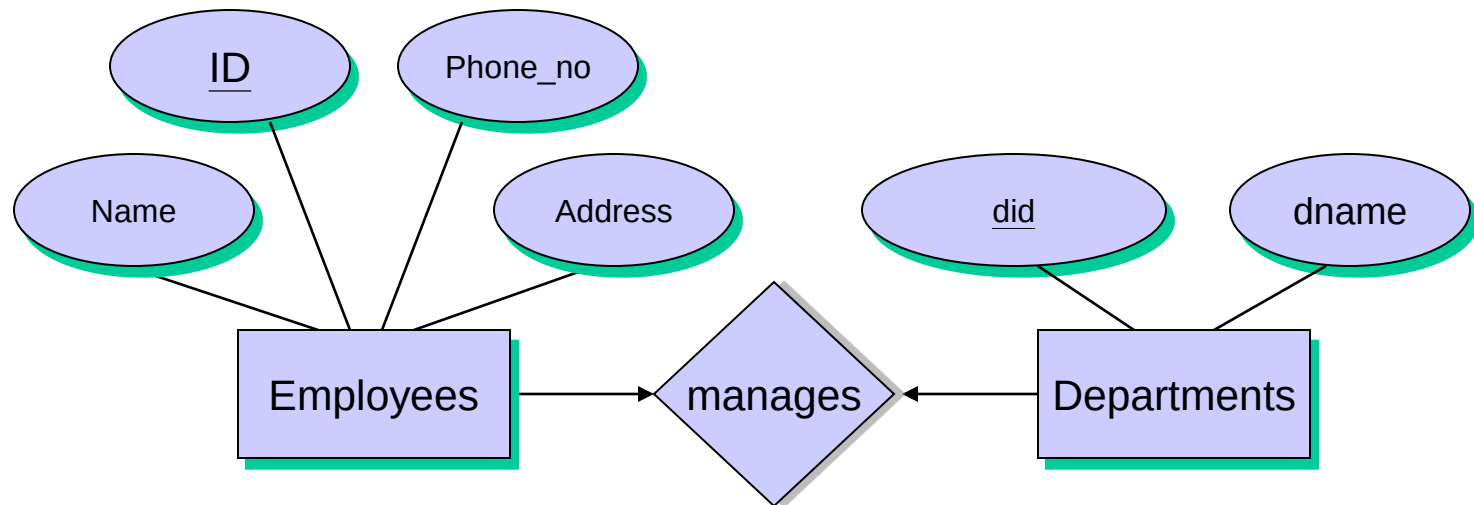
Notation: Many-to-Many/ $m:n$



- A customer can associate with several loans (**possibly 0**) via Borrower
- A loan can associate with several customers (**possibly 0**) via Borrower
- **(Notice: we do not assume all entity instances to participate in the relationships they are in)**

Limitation of Cardinality Alone

- With reference to below diagram:
 - Constraint on Manages tells us that one department has **at most** one manager.



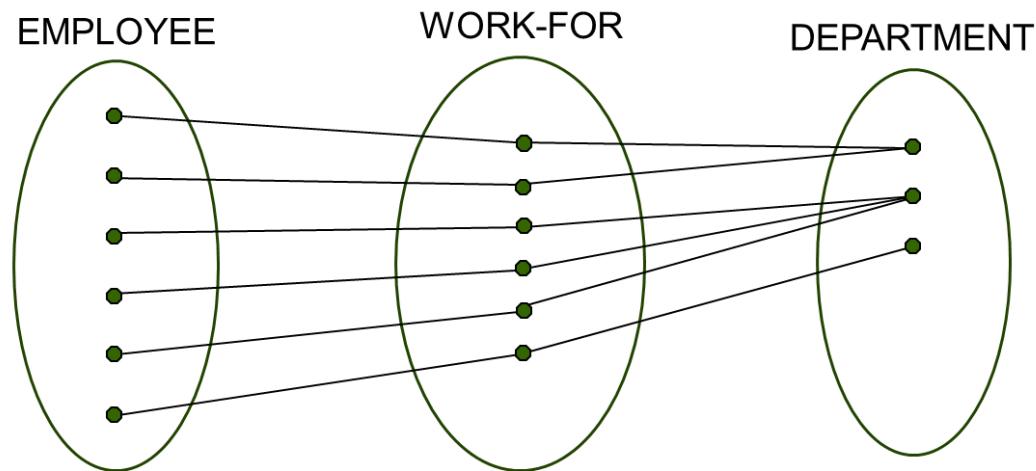
- Does this currently express whether every department is required to have a manager?

Participation Constraints

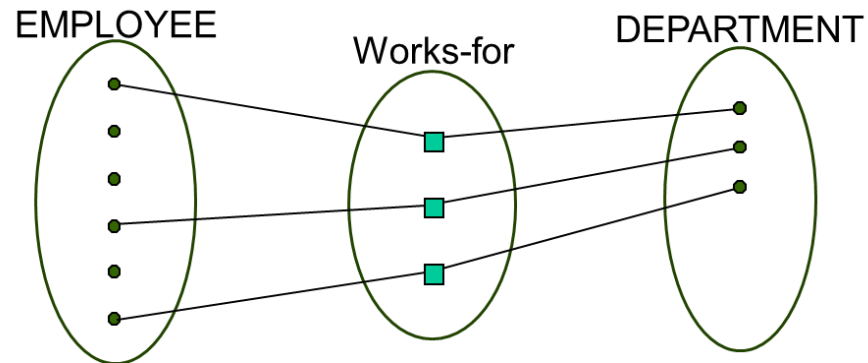
- Example: Course and Students (rel. = enrolment)
 - students must be enrolled in a course?
 - course must have students enrolled?
- We can classify participation in relationships as follows:
 - **total** - each entity in the entity set must participate in at least one relationship.
 - **partial** - an entity in the entity set may not participate in a relationship.
- More generally, will all instances of this entity participate in this relationship? (see next slide for example visualized)

Participation Visualized

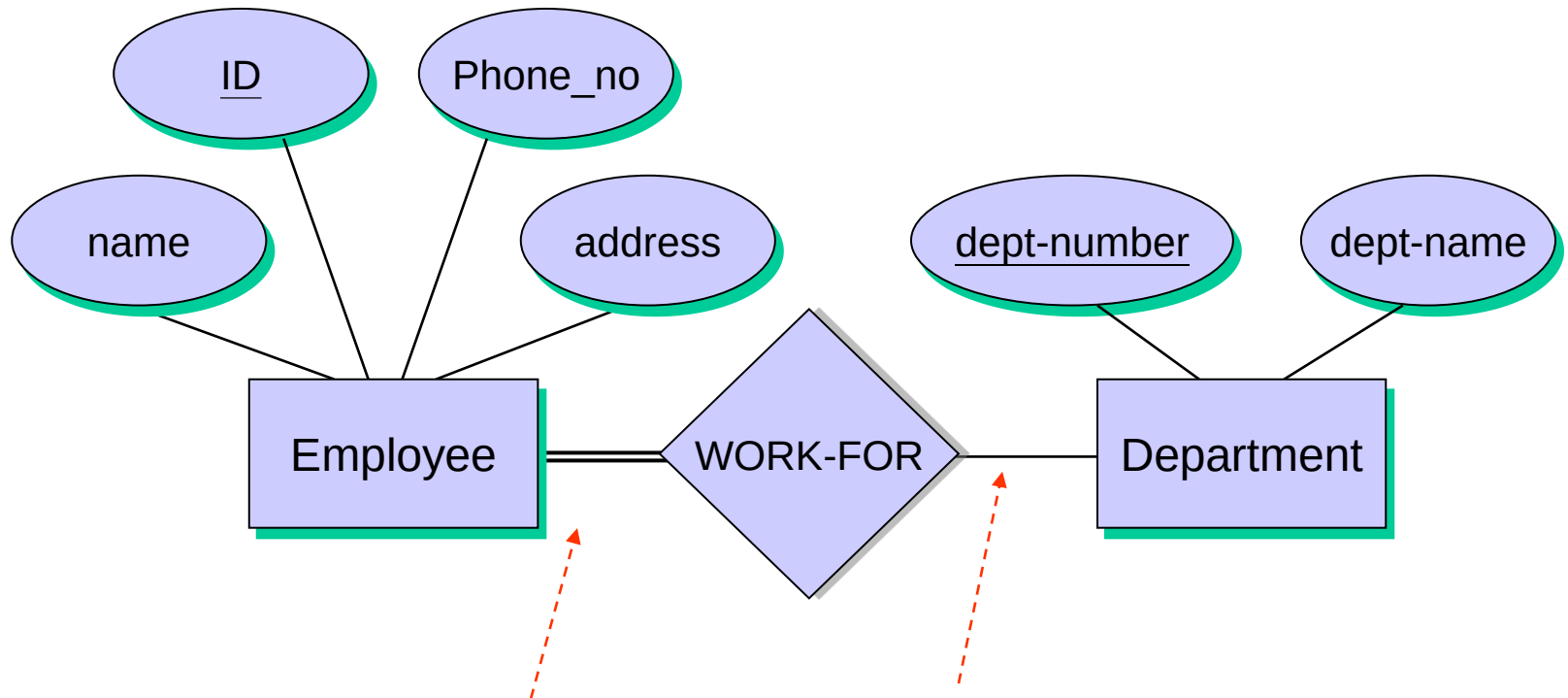
- Every employee must work for some department. The participation of EMPLOYEE in WORKS-FOR is total participation.



- Partial participation counter example ->



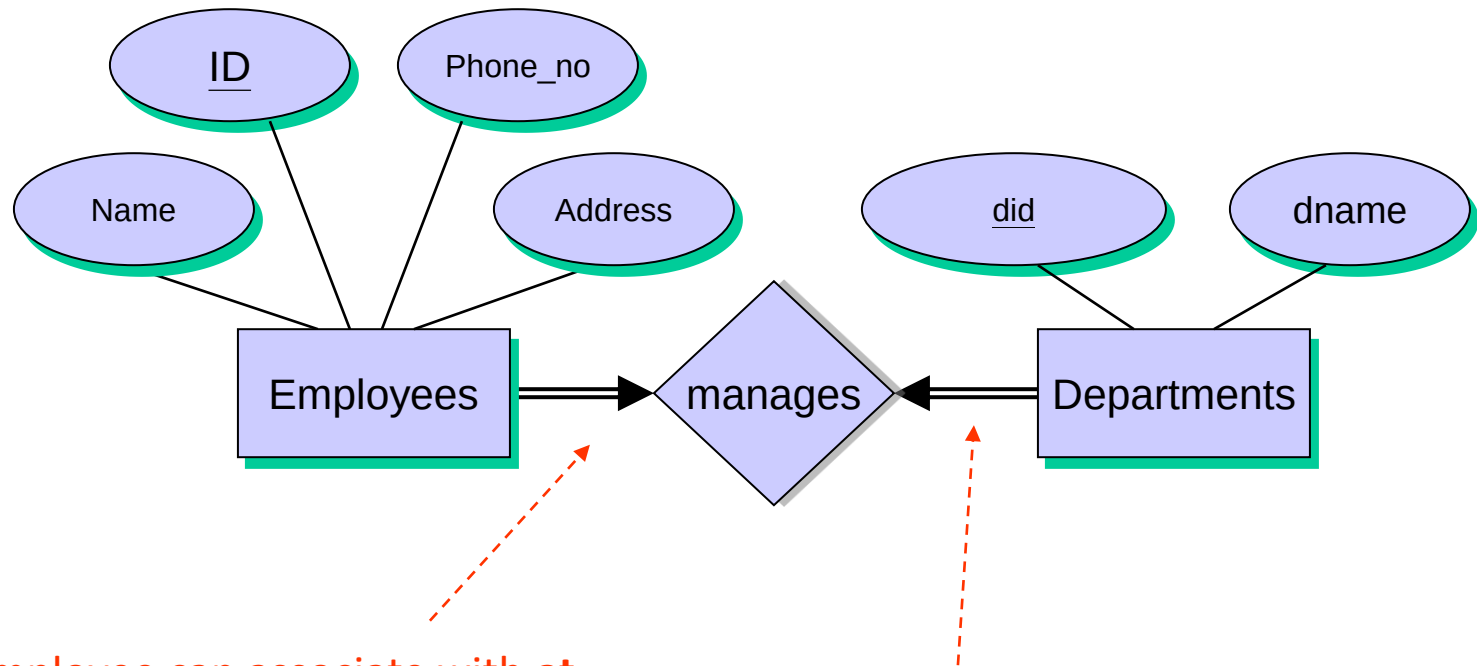
Notation: Participation Constraint



If the participation of an entity in relationship is total, two are connected by a **double link**.
If the participation of an entity in relationship is partial, two are connected by a **single link**.

In this case, the thick line means every employee has to participate in the work-for relationship.
i.e., every employee has to work for some departments (at least one department).

Note: Combined Implications



An employee can associate with **at most** one department via the relationship “manages”?

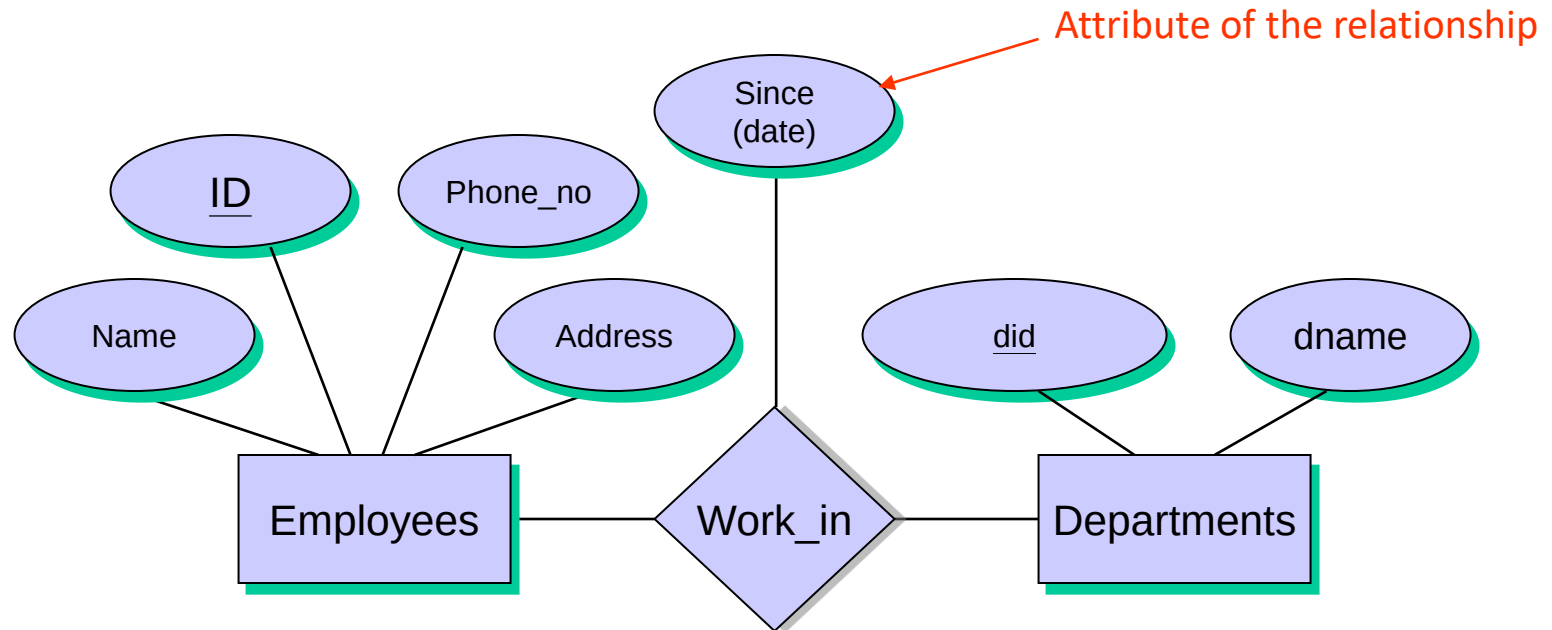
An employee can associate with **at least** one department via the relationship “manages”?

A department can associate with **at most** one employee via the relationship “manages”?

A department can associate with **at least** one employee via the relationship “manages”?

Relationship Attributes*

- A relationship can also have descriptive attributes.
- **Descriptive attributes** are used to record information about the relationship, rather than about any one of the participating entities.

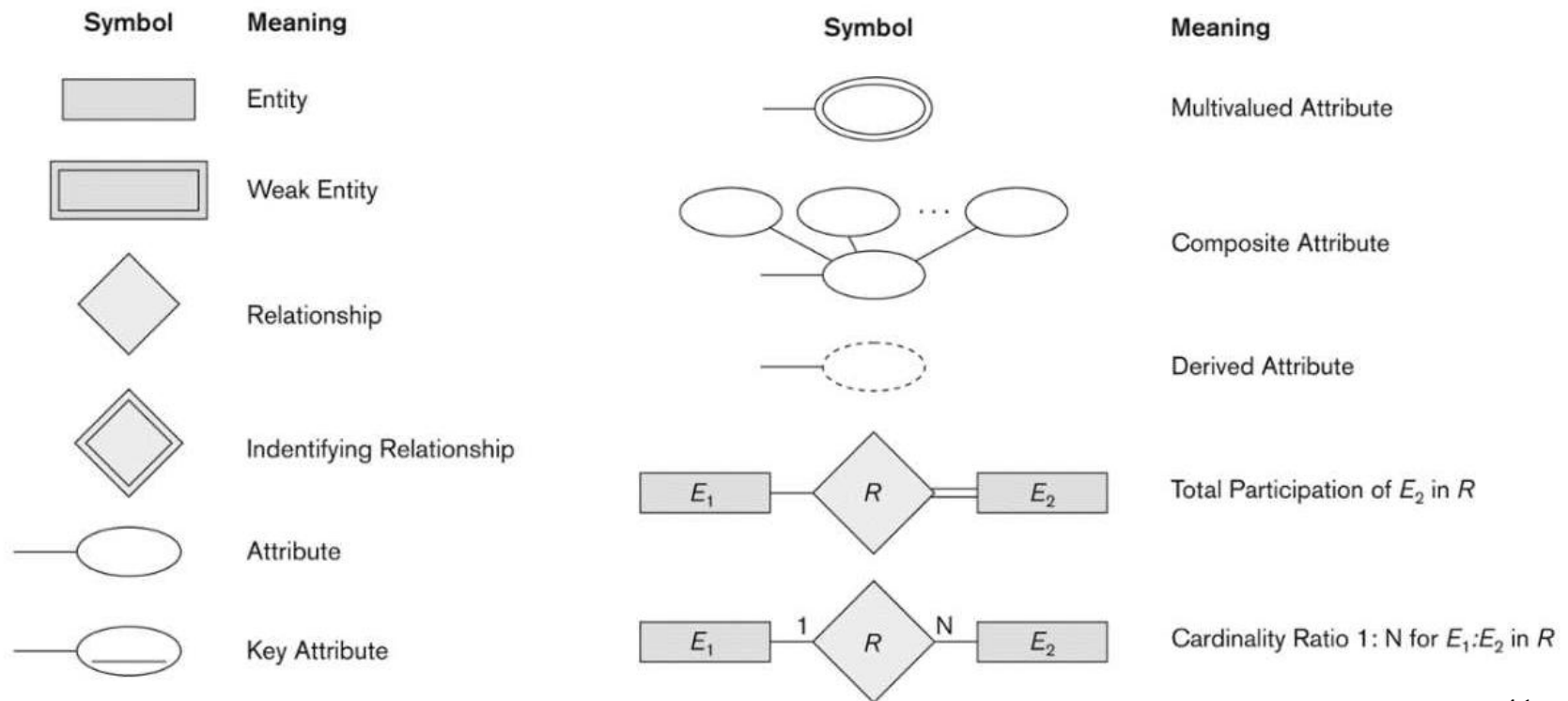


Thoughts On Database Design with ER

- Developing an ER diagram presents several **choices**,
 1. *Should a concept be modeled as an entity or an attribute?*
 2. *Should a concept be modeled as an entity or a relationship?*
 3. *What relationship? Which entities to involve?*

Important: Course Unique Notation

- Our Notation is largely based on **Chen's notation**
 - **HOWEVER! Only cardinality is expressed differently!**
 - **Be mindful of other similarly looking notations! They will not mean the same thing always.**



Check Understanding

- Not the course notation! Similar yes.
 - Please treat thickened lines as double lines

